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OpenDGM: A modular Python framework for electroanatomical data analysis and arrhythmia mechanism detection

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Software

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Summary

OpenDGM is a modular Python framework designed to parse, analyse, and visualise simulated, clinical and experimental electrophysiological data. The software aims to lower the barrier to entry for researchers and clinicians working in cardiac electrophysiology by providing reusable tools for data processing, analysis, and visualisation.

OpenDGM supports electrophysiological datasets from multiple clinical mapping systems, simulation platforms, and optical mapping experiments. Its modular design allows users to rapidly construct pipelines for quantitative analysis and visualisation without the need for extensive software development.

Beyond data analysis, OpenDGM provides automated algorithms for detecting and characterising the underlying mechanisms of arrhythmias. These tools facilitate the interpretation of clinical recordings and the validation of computational models, helping users gain mechanistic insights into complex electrophysiological phenomena, such as rotational activity, focal sources and phase singularities.

By combining accessibility, flexibility and advanced analysis capabilities in a modular framework, OpenDGM aims to accelerate methodological development and reproducible research in cardiac electrophysiology.

Statement of need

Electroanatomical tools for cardiac ablation therapy guidance are central to contemporary electrophysiology practice and research; however, widely used systems such as CARTO (Biosense Webster, Inc., USA), EnSite (Abbott Laboratories, USA), Rhythmia (Boston Scientific Corporation, USA), and many new market entrants are proprietary software. As a result, their underlying methods are not fully transparent in the scientific literature, limiting reproducibility and hindering the development and comparison of new analysis approaches.

OpenDGM addresses the need for open and extensible software for working with electroanatomical data, enabling reproducible analysis and the development and validation of new computational methods on both simulated and clinical datasets.

The development of OpenDGM was motivated by research on experimentally induced Torsade de Pointes, which initially led to a closed-source C++ software package DGM (Van Den Abeele et al., 2023; Vandersickel et al., 2017). As subsequent research projects required analysis of increasingly diverse electroanatomical datasets, OpenDGM was developed to avoid repeated

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reimplementation of common functionality and to consolidate it into a reusable, extensible, and open-source Python framework.

State of the field

To our knowledge, no existing software packages provide the automated arrhythmia mechanism detection algorithms implemented in OpenDGM.

Software design

OpenDGM is designed as a modular framework in Python. This enables users to construct flexible analysis pipelines in which individual components can easily be replaced or combined to evaluate different methods and configurations.

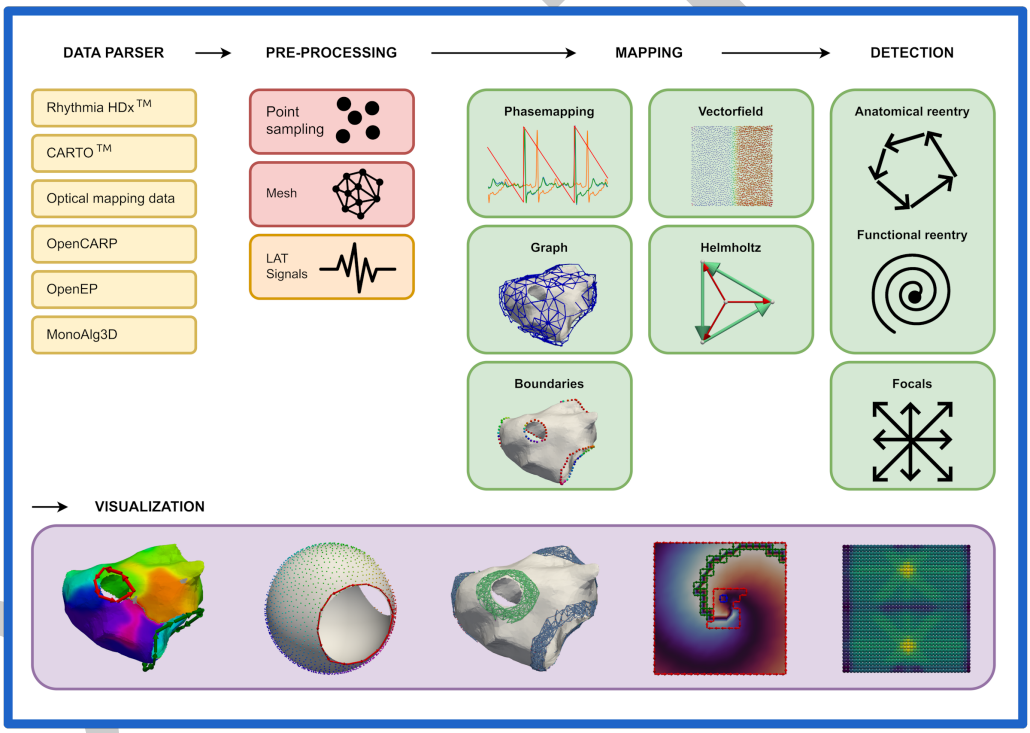


Figure 1: Overview of how OpenDGM modules can be combined into a modular analysis pipeline.

Modules

The framework consists of the following core modules, which can be combined as illustrated in (Figure 1):

Module	Task
core	The core module contains internal helpers used by the other modules, such as validation decorators, a method timer and logging.
parser	Parsing of electrophysiological mapping data from clinical mapping systems, simulation software, and optical mapping experiments, including CARTO™, Rhythmia HDx™, OpenEP, OpenCARP, and MonoAlg3D (EnSight).
signal	Module for signal processing. Multiple predefined signal filters are implemented, as well as methods to annotate electrograms and extract local activation times (LATs) or phases.
topo	The topo module provides the topological objects on which all algorithms operate. The central object is a <code>Complex</code>, which represents the simplex layers of a surface or volumetric mesh (points, edges, faces, and cells) together with the data defined on them. It serves as the primary data structure for inputs and outputs across the framework. A toolkit is provided to construct, manipulate, and preprocess these objects, for example by extracting subcomplexes.
algorithm	The algorithm module contains computational methods using topology- and network-based approaches. Algorithms are provided to detect phase singularities, reentry cycles, focal sources and sinks, and boundaries around which reentrant activity can anchor.
visualization	Lightweight wrapper around PyVista providing additional functionality, including animated time-series visualisation.

By combining these modules, users can construct custom pipelines for parsing, analysing, and visualising electrophysiological data.

Common use cases

OpenDGM supports a range of workflows in cardiac electrophysiology research, including:

- **Detection of rotational activity**: Detect phase singularities and reentry cycles in electrophysiological data, both anatomical and functional.
- **Data parsing**: Load and parse simulated, clinical and experimental mapping data.
- **Exploring mesh topology**: Extract features from the mesh and calculate quantities such as phase defects and topological charges.
- **Identifying network structures**: Use network-based methods to identify focal sources and sinks and detect cycles in the underlying electrophysiological network.
- **Signal analysis**: Filter electrograms, annotate local activation times and calculate phase maps.
- **Visualisation of electrophysiological phenomena**: Visualise electroanatomical data and spatial structures such as wavefronts and boundaries, including their evolution over time.
- **Development of custom methods**: Combine OpenDGM's data structures and existing functionality with user-defined algorithms to implement and evaluate new analysis approaches.

Research impact statement

OpenDGM has been developed within the Biophysics Research Group at Ghent University and has been used in multiple peer-reviewed studies involving simulated, clinical and experimental electrophysiological data. A full list of publications and collaborations using OpenDGM is available at https://biophysics.ugent.be.

Representative applications include methods for detecting rotational activity in cardiac

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arrhythmias using topology- and network-based approaches (Lootens et al., 2025; Van Den Abeele, Lootens, et al., 2025; Van Den Abeele, Hendrickx, et al., 2025), studies applying OpenDGM to clinical electroanatomical mapping data (Duytschaever et al., 2024, 2025; Hawson et al., 2023) and simulation-based investigations of arrhythmia mechanisms (Bezerra et al., 2025; Verstraeten et al., 2026).

Code availability

The source code and examples are available at https://gitlab.com/opendgm/opendgm.

API reference documentation is available at https://opendgm.ugent.be.

AI usage disclosure

Generative AI tools were used occasionally during development to generate or rewrite small code snippets, such as simple loops and helper functions, and to improve the wording of documentation and manuscript text.

Generative AI was not used to make architectural or software design decisions.

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